

Three-Dimensional Position Reconstruction in the API System

*Mathematical derivation and reconciliation of the `calcXYZ`, `api_xyz`,
and thesis (Sec. 4.3) formulations*

This note derives, from first principles and for our specific geometry, the reconstruction of the neutron-interaction point (x, y, z) in the Associated-Particle Imaging (API) system. The full model (relativistic speeds, center-of-mass correction, exact gamma-detector vector) is taken as authoritative; the two simplified implementations and the thesis derivation are recovered as special cases, and every discrepancy among the three sources is catalogued in the errata of Section 8.

1. The reconstruction problem

In coincidence mode the DAQ reports, per event, the four corner energies of the YAP alpha detector, the gamma-ray energy, and the alpha–gamma time difference dt (Table 4.2 of the thesis). Reconstruction proceeds in four steps:

- **Step A — Alpha position.** The four corner energies give the alpha interaction point (x_0, y_0) on the YAP face (Section 3).
- **Step B — Neutron direction and speed.** Momentum conservation in the D–T reaction fixes the neutron velocity vector from the alpha direction, including the center-of-mass motion (Section 4).
- **Step C — Time of flight.** The measured dt , the alpha flight time t_α , and the gamma flight time tie the neutron path length to the geometry (Section 5).
- **Step D — Solve for the point.** A single quadratic in the neutron flight time yields (x, y, z) (Section 6).

The tagged neutron travels essentially opposite to the detected alpha, so the alpha hit defines a *ray* from the source into the sample; the timing fixes *how far* along that ray the gamma-producing scatter occurred.

2. Coordinate system and geometry

We place the origin O at the neutron production point (the reaction spot inside the generator head) and let the $+z$ axis point along the API cone axis, into the sample volume. This is the choice made by `calcXYZ`; it is the cleanest because the neutron starts at the origin, so both the interaction point and the gamma detector are referred to the same point. The YAP alpha-detector face lies at $z = -z_t$, and the alpha interacts at $A = (x_0, y_0, -z_t)$.

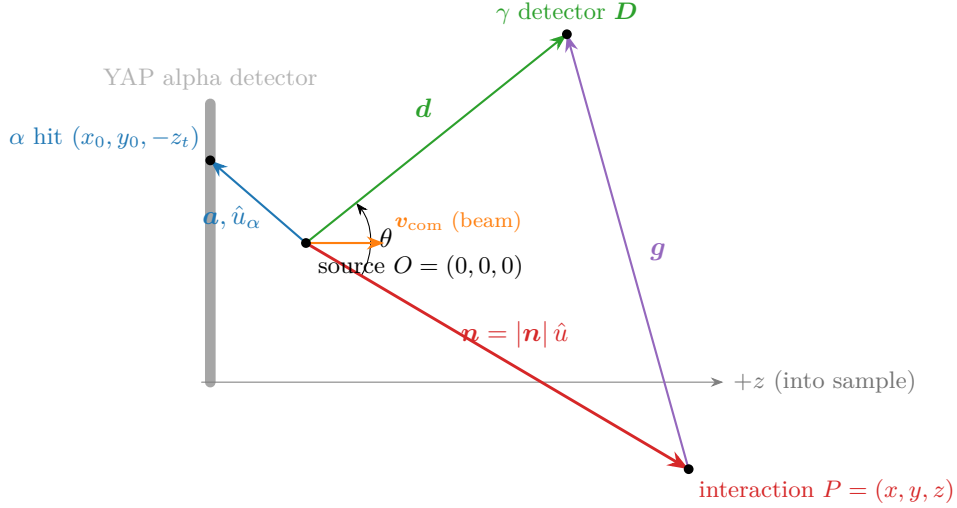


Figure 1. Reconstruction geometry (projected on the y - z plane). The alpha hit on the YAP defines the ray direction; the timing fixes the neutron path length $|n|$. Vectors $\mathbf{a}, \mathbf{n}, \mathbf{d}, \mathbf{g}$ and the angle θ are used in the derivation.

Our specific geometry uses the following fixed quantities:

Symbol	Meaning	Value (this system)
z_t	source $\rightarrow$ YAP-face distance	6.7 cm
a	YAP active square side	4.8 cm (± 2.4 cm)
v_α	alpha speed (3.5 MeV, relativistic)	$1.298 \times 10^7 \text{ m s}^{-1}$ ($\beta = 0.0433$)
v_n	neutron speed (14.1 MeV, relativistic)	$5.136 \times 10^7 \text{ m s}^{-1}$ ($\beta = 0.1713$)
v_{com}	center-of-mass speed (50 keV D^+)	$8.764 \times 10^5 \text{ m s}^{-1}$ ($\beta = 0.00292$)
c	speed of light = v_γ	$2.997\,925 \times 10^8 \text{ m s}^{-1}$
$\hat{C}$	ion-beam unit vector (v_{com} dir.)	<code>meta['ion beam axis']</code>
D	gamma-detector centre	<code>meta['detectors'][k]['position']</code>

In `calcXYZ` the symbol `meta['scintillator distance']` is z_t : it is the z of the alpha hit, `Z_alpha = scintillator_dz`. The thesis quotes $z_t = 6$ cm in the text; the code uses 6.7 cm, which we adopt here (see errata, Section 8).

3. Step A — alpha position from the four corners

The YAP is read out at its four corners A, B, C, D (charge division). Labelling them so that B is the $+x, +y$ corner, A the $-x, +y$, C the $+x, -y$, and D the $-x, -y$ corner, the normalised coordinates are

$$x_{\text{raw}} = \frac{B + C - D - A}{A + B + C + D}, \quad y_{\text{raw}} = \kappa \frac{B + A - D - C}{A + B + C + D}, \quad (1)$$

where $\kappa = 1.305$ is an empirical factor that makes the measured alpha image square (`map_alpha_XY`). Note `calc_own_pos` / the `x2, y2` columns used by the GUI omit κ (set it to 1) and apply the squaring later through edge detection — numerically equivalent once the axis is scaled to centimetres.

The dimensionless values are mapped to physical centimetres across the 4.8 cm active face. Rather than trust the absolute gain, the code locates the populated edges of the alpha image (the histogram region above $\sim 1/3$ of the peak) and linearly maps that span onto $[-2.4, +2.4]$ cm:

$$x_0 = \frac{a}{x_{\max} - x_{\min}} (x_{\text{raw}} - x_{\max}) + \frac{a}{2}, \quad a = 4.8 \text{ cm}, \quad (2)$$

and similarly for y_0 . (The legacy `map_alpha_XY` additionally flips the x -axis so an X - Y image matches the view from in front of the experiment, and can apply spline edge-corrections from the `DX/DY` lookup tables.) The output of Step A is the alpha interaction point (x_0, y_0) in centimetres on the YAP face.

4. Step B — neutron velocity vector

4.1 Naive (back-to-back) direction

If the neutron and alpha left the source exactly 180° apart, the neutron direction is simply opposite the source $\rightarrow$ alpha vector. With the alpha hit at $A = (x_0, y_0, -z_t)$,

$$\mathbf{a} = \langle x_0, y_0, -z_t \rangle, \quad \hat{u} = -\frac{\mathbf{a}}{|\mathbf{a}|} = \frac{\langle -x_0, -y_0, z_t \rangle}{\sqrt{x_0^2 + y_0^2 + z_t^2}}. \quad (3)$$

This is exactly the thesis vector (eq. 4.5) and what `api_xyz` uses ($u_x = -x_a/|\mathbf{a}|$, $u_z = +z_a/|\mathbf{a}|$ with $z_a = 6.7$ cm). The thesis takes the origin at the YAP centre rather than the source, but the *direction* $\hat{u}$ is identical.

4.2 Center-of-mass correction (the full model)

The 180° assumption holds only in the reaction center-of-mass (CM) frame. The ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He} + n$ reaction is driven by a deuteron beam, so the CM moves in the lab with a small velocity along the beam axis $\hat{C}$:

$$\mathbf{v}_{\text{com}} = v_{\text{com}} \hat{C}, \quad v_{\text{com}} = \frac{\sqrt{2 m_d E_d}}{m_d + m_t} c \approx 0.00292 c. \quad (4)$$

For a 50 keV deuteron beam this is $v_{\text{com}} \approx 8.764 \times 10^5 \text{ m s}^{-1}$. Although tiny ($\beta_{\text{com}} \approx 0.00292$), it tilts the lab neutron direction away from exact 180° by up to $\approx 0.98^\circ$, i.e. roughly 1.2 cm of transverse displacement at a 70 cm stand-off — comparable to the system resolution and therefore worth correcting. `calcXYZ` reads v_{com} versus beam energy from `center-of-mass.npy`; the formula above reproduces it from the reaction kinematics.

The construction has three parts. First the alpha flight time t_α is found by requiring that, in the CM frame, the alpha travels at speed v_α :

$$|\mathbf{A} - \mathbf{v}_{\text{com}} t_\alpha| = v_\alpha t_\alpha \implies (v_\alpha^2 - v_{\text{com}}^2) t_\alpha^2 + 2(\mathbf{A} \cdot \mathbf{v}_{\text{com}}) t_\alpha - |\mathbf{A}|^2 = 0, \quad (5)$$

taking the positive root. The alpha velocity in the CM frame is then $\mathbf{w}_\alpha = (\mathbf{A} - \mathbf{v}_{\text{com}} t_\alpha)/t_\alpha$, and because the neutron is exactly opposite the alpha *in the CM frame*, the neutron velocity in the lab is

$$\mathbf{v}_n = -\frac{\mathbf{w}_\alpha}{v_\alpha} v_n + \mathbf{v}_{\text{com}}, \quad (6)$$

i.e. (neutron CM direction) $\times$ (neutron CM speed) + (CM drift). This is exactly the two-line update in `calcXYZ` (`vn_x = -vn_x/v_alpha*v_neutron + v_com_vector[0], ...`). In the naive model $v_{\text{com}} \rightarrow 0$ and $\mathbf{v}_n = v_n \hat{u}$.

5. Step C — the time-of-flight relation

The recorded dt is the alpha–gamma time difference. Writing t_n, t_g, t_α for the neutron, gamma, and alpha flight times, the thesis relation (eq. 4.7) is

$$dt = t_n + t_g - t_\alpha \implies dt_a \equiv dt + t_\alpha = t_n + t_g, \quad (7)$$

so folding the alpha flight time into the time difference defines a single quantity dt_a . (Per-detector electronic offsets — the `z-align` constants and the ≈ 11 ns PMT transit delay noted in the thesis — are subtracted from dt before this step; they shift the zero of dt but do not change the algebra below.) In `calcXYZ` this is literally `df['dt'] = dt_orig + t_alpha` followed by the channel-wise offset subtraction.

6. Step D — solving for the interaction point

Let $\mathbf{D}$ be the gamma-detector centre and $\mathbf{P} = \mathbf{v}_n t_n$ the interaction point (the neutron starts at the origin with lab velocity $\mathbf{v}_n$). The gamma then travels from $\mathbf{P}$ to $\mathbf{D}$ at speed c in the remaining time $t_g = dt_a - t_n$:

$$|\mathbf{D} - \mathbf{v}_n t_n|^2 = c^2 (dt_a - t_n)^2. \quad (8)$$

Expanding gives a quadratic in the neutron flight time, $\alpha' t_n^2 + \beta' t_n + \gamma' = 0$, with

$$\alpha' = c^2 - |\mathbf{v}_n|^2, \quad \beta' = -(2c^2 dt_a - 2\mathbf{D} \cdot \mathbf{v}_n), \quad \gamma' = c^2 dt_a^2 - |\mathbf{D}|^2. \quad (9)$$

Our coordinate choice selects the negative root:

$$t_n = \frac{-\beta' - \sqrt{\beta'^2 - 4\alpha'\gamma'}}{2\alpha'}, \quad (x, y, z) = \mathbf{v}_n t_n. \quad (10)$$

This is exactly the second quadratic in `calcXYZ` (coefficients `a, b, c` and `t_neutron`); the reconstructed point is `X=vn_x*t_neutron`, etc. It uses the *exact* detector vector $\mathbf{D} - \mathbf{P}$; no point-source or law-of-cosines approximation enters.

6.1 The thesis / `api_xyz` special case

If we drop the CM correction (so $|\mathbf{v}_n| = v_n$ and $\mathbf{v}_n = v_n \hat{u}$) and solve instead for the path length $|\mathbf{n}| = v_n t_n$, the gamma constraint becomes the law of cosines (thesis eq. 4.8)

$$|g|^2 = |\mathbf{d}|^2 + |\mathbf{n}|^2 - 2|\mathbf{d}||\mathbf{n}|\cos\theta, \quad \cos\theta = \frac{u_x x_1 + u_y y_1 + u_z z_1}{|\mathbf{d}|}, \quad (11)$$

which, combined with $|g| = c(dt_a - |\mathbf{n}|/v_n)$, yields the thesis quadratic (eqs. 4.9–4.10)

$$a = 1 - \frac{c^2}{v_n^2}, \quad b = \frac{2dt_a c^2}{v_n} - 2|\mathbf{d}|\cos\theta, \quad c = |\mathbf{d}|^2 - c^2 dt_a^2, \quad (12)$$

$$|\mathbf{n}| = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \quad x = \frac{-x_0}{|\mathbf{a}|}|\mathbf{n}|, \quad y = \frac{-y_0}{|\mathbf{a}|}|\mathbf{n}|, \quad z = \frac{z_t}{|\mathbf{a}|}|\mathbf{n}| + z_t. \quad (13)$$

This is precisely `api_xyz(use_det=True)`. The two forms agree when $v_{\text{com}} \rightarrow 0$ and when $\mathbf{d}$ is measured from the same point as $\mathbf{n}$ — which exposes the origin inconsistency discussed in the errata. Setting `use_det=False` drops the detector term entirely, leaving the straight-ray estimate $|\mathbf{n}| = v_n dt_a$; this is the crudest variant and is, at present, the one the `--beta` GUI uses.

7. Reconciliation of the three implementations

All three sources implement the same idea at increasing levels of rigour.

Aspect	Thesis Sec. 4.3	api_xyz (--beta)	calcXYZ
Coordinate origin	YAP centre	YAP centre	neutron source
Source–YAP z_t	6 cm	6.7 cm	meta scint. dist.
Particle speeds	symbolic	non-relativistic	relativistic
α – n back-to-back	exact 180°	exact 180°	CM-corrected
Gamma detector	law of cosines	cosines / ignored	exact $\mathbf{D} - \mathbf{P}$
Units	—	cm, $c = 3 \times 10^{10}$	SI, scipy c
Solves for	$ \mathbf{n} $ (path)	$ \mathbf{n} $ (path)	t_n (then $\mathbf{v}_n t_n$)
Quadratic root	negative	negative	negative

8. Errata and resolved discrepancies

The following issues were found while reconciling the three sources. Items 1–4 are genuine physics/algebra points; items 5–6 are bookkeeping.

- Thesis typo (minor).** The constant term of eq. 4.10 is printed as $c = |\mathbf{d}|^2 - c^2 dt^2$ but should use dt_a , i.e. $c = |\mathbf{d}|^2 - c^2 dt_a^2$. (The stray subscript “ a ” is dropped to the next line in the PDF.) The rest of the derivation is self-consistent.
- Law-of-cosines origin mismatch (thesis & api_xyz).** The neutron path $\mathbf{n}$ is measured from the source at $(0, 0, z_t)$, but $\mathbf{d}$ and $\cos \theta$ are measured from the YAP centre. The law of cosines treats them as sharing one vertex, so the gamma path length is off by $\sim z_t$ in z . Putting the origin at the source (full model) removes this: $\mathbf{D}$ and $\mathbf{P}$ share the origin and $|\mathbf{g}| = |\mathbf{D} - \mathbf{P}|$ is exact.
- Non-relativistic speeds (api_xyz).** Using $\sqrt{2E/m}$ overestimates the neutron speed: $\beta_n = 0.1732$ vs 0.1713 relativistic, $\sim 1.1\%$ high. Over a ~ 70 cm flight this biases Z at the centimetre scale. The full model uses the relativistic β .
- 180° back-to-back assumption (thesis & api_xyz).** Ignores the lab-frame CM momentum; the neutron is back-to-back with the alpha only in the CM frame. This tilts the ray by up to $\sim 0.98^\circ$ (~ 1.2 cm at 70 cm). The full model corrects it via $\mathbf{v}_{\text{com}}$ — precisely the dominant error source the thesis itself flags.
- z_t value.** 6 cm (thesis text) vs 6.7 cm (code). We adopt 6.7 cm, matching `api_xyz`, `helper_api`, and the as-built code.
- Sign / origin convention.** `api_xyz` puts the sample at $+z$ with the YAP centre as origin; `calcXYZ` puts the origin at the source. They are related by a translation of z_t along z (and a sign flip of the z -axis). Reconstructed depths must be interpreted in the convention of whichever routine produced them.

Practical consequence for --beta. The GUI currently calls `api_xyz(use_det=False)`, the crudest path (non-relativistic, no CM correction, detector ignored). Migrating the GUI to the full `calcXYZ` model — or at minimum enabling the detector term and relativistic speeds — is the natural follow-up to this note.

9. Worked numerical example

To make the formulas checkable, we reconstruct one synthetic event with the full model. Inputs (origin at source, $+z$ into the sample):

- alpha hit on the YAP: $(x_0, y_0) = (1.20, -0.80)$ cm, at $z = -6.7$ cm;
- measured time difference: $dt = 18.0$ ns;
- beam axis $\hat{C} = (0, 0, 1)$; $v_{\text{com}} = 8.764 \times 10^5$ m s $^{-1}$;
- illustrative gamma-detector centre: $\mathbf{D} = (0, 30, 35)$ cm.

Intermediate and final results:

Quantity	Value
alpha flight time t_α	5.653 ns
$dt_a = dt + t_\alpha$	23.653 ns
neutron flight time t_n (full)	21.075 ns
$\mathbf{P}$, full model (det + CM + rel.)	(−17.70, 11.80, 107.98) cm
$\mathbf{P}$, detector ignored (--beta path)	(−19.54, 13.03, 119.19) cm
$\mathbf{P}$, thesis (non-rel, no CM)	(−18.89, 12.59, 105.47) cm
depth difference full – thesis (z)	2.51 cm

The three models agree to within a few centimetres on this event, with the differences dominated by the detector term and the CM/relativistic corrections — consistent with the system’s few-centimetre resolution and confirming that the simplified routines are adequate for quick looks while the full model is preferred for quantitative work.

Numbers above are produced by `scratch/build_api_reconstruction_doc.py`, which re-implements the three models directly; the detector position is illustrative.